Statistical Analysis Report

Bank Marketing Campaign & Customer Response Analytics ·

1. Approach

This report tests whether the patterns found in the exploratory analysis (Sprint 2) are statistically supported, measures how precisely key figures are known, and evaluates the engineered features. Rules followed throughout:

- Significance level $\alpha = 0.05$ for every test.
- With 11,162 customers almost any difference is 'significant', so every test also reports an **effect size** (Cramér's V, Cohen's d, probability of superiority, η^2).
- Balance, duration and number of contacts are heavily right-skewed, so rank tests (Mann–Whitney, Kruskal–Wallis) are the primary tests; Welch t and ANOVA are reported for comparison.
- The 14 chi-square tests are also checked against a Bonferroni-adjusted threshold ($0.05 / 14 = 0.0036$).
- The patterns were first spotted on the same data, so p-values are somewhat optimistic; feature evaluation therefore uses held-out data.

2. Descriptive statistics

Variable	Median	Skewness	Excess kurtosis	Outliers beyond 1.5×IQR
Age	39	0.88	0.68	1.5%
Balance	551	8.28	128	9.5%
Call duration (s)	255	2.17	7.48	5.9%
Contacts in campaign	2	5.55	57.4	5.4%
Previous contacts	0	7.34	106	11.3%

- Balance, contacts and previous contacts have very heavy tails, so medians and IQR (balance IQR 1,566; age IQR 17) describe the typical customer better than means (balance mean 1,520 vs median 551).
- For pdays the IQR rule is meaningless because the code −1 (never contacted) dominates; the 'previously contacted' flag is used instead.

- Spearman correlation with subscribing: duration +0.52, previous contacts +0.23, pdays +0.21, balance +0.16, number of contacts −0.13, age −0.01 (p = 0.24, not significant). All others have p < 1e-40.
- Covariances are unit-dependent: balance moves with age (+4,183) and with duration (+25,788); number of contacts moves slightly against duration (−39). Strength is read from correlation, not covariance.

3. Hypothesis tests

3.1 Chi-square: is a categorical factor associated with subscribing?

H0: the factor and the deposit outcome are independent. H1: they are associated.

Factor	χ^2	df	p-value	Cramér's V	Decision
Contact method	693.5	2	2.5e-151	0.25	Reject H0
Previous campaign outcome	991.3	3	1.4e-214	0.30	Reject H0
Housing loan	464.0	1	6.5e-103	0.20	Reject H0

All expected cell counts are well above 5. Eleven further factors were tested the same way: **all 14 reject H0 and all survive the Bonferroni threshold**. Strongest effects: month (V = 0.31), previous outcome (0.30), contact method (0.25), debt status (0.23), previously contacted (0.23). Marital status (0.10) and default (0.04) are significant but their effect is negligible.

Business meaning: campaign history, contact channel, debt load and timing are worth building call strategy around; default and marital status can be ignored for targeting. Month is the strongest association but is entangled with volume and contact method, so it should not be used alone to move budget.

3.2 Subscribers vs non-subscribers: do numeric characteristics differ?

H0: the variable has the same distribution in both groups. H1: it differs. Primary test: Mann–Whitney.

Variable	Median (yes vs no)	p-value	Effect size	Decision
Call duration (s)	418 vs 165	< 1e-300	d = 0.99; P(superiority) 0.80	Reject H0 (very large)
Account balance	718 vs 431	3.1e-61	d = 0.16; P = 0.59	Reject H0 (small)

Variable	Median (yes vs no)	p-value	Effect size	Decision
Age	38 vs 39	0.24	d = 0.07	Fail to reject H0
Contacts in campaign	2 vs 2 (means 2.1 vs 2.8)	9.3e-41	d = −0.26; P = 0.43	Reject H0 (small)

Bootstrap 95% CI for the median difference: duration 243–269 seconds; balance 239–347. For age the Welch t-test is 'significant' (p = 0.0002) but the difference is under one year and the rank test disagrees; age matters only through its U-shape, which neither test captures (hence the age bands in section 5).

Business meaning: call quality is the only strong numeric separator, but it is known only after the call. Wealthier customers respond somewhat more; subscribers received fewer calls on average because heavily-called customers rarely convert.

3.3 Three or more groups

H0: the groups are the same. H1: at least one differs. Primary test: Kruskal–Wallis.

Question	Statistic	p-value	Effect / comparison	Decision
Balance across 12 job groups	H = 192.2	3.0e-35	η^2 = 0.012 (ANOVA F = 12.6, p = 3.9e-24)	Reject H0, but job explains ~1% of balance variation
Call duration across contact methods	H = 55.9	7.2e-13	Medians 262 / 245 / 229 s; $\eta^2 \approx 0$ (ANOVA p = 0.084 disagrees)	Reject H0, but only ~30 s apart
Campaign ROI across 5 marketing channels (83 campaigns)	H = 2.0	0.73	ANOVA F = 0.32, p = 0.86	**Fail to reject H0**

Business meaning: job is a weak proxy for wealth. Call length does not explain channel differences in response. Most important, **the real marketing table does not show that channels differ in ROI**, so budgets should not be reallocated on the Sprint 2 ranking alone.

4. Confidence intervals (95%)

Metric	Estimate	95% CI
Overall response rate	47.4%	46.5 – 48.3
Previous campaign success	91.3%	89.5 – 92.9
No previous history	40.8%	39.8 – 41.9
No housing loan	57.0%	55.8 – 58.3
Housing loan	36.6%	35.4 – 37.9
Cellular contact	54.2%	53.2 – 55.3
Unknown contact method	23.8%	22.2 – 25.6
1 call	53.4%	52.0 – 54.8
6+ calls	29.3%	26.3 – 32.4
Debt-free, age 60+	79.4%	76.1 – 82.3
Debt-free, under 30	75.9%	72.8 – 78.8
Blue-collar with housing loan	32.7%	30.2 – 35.3
Overdrawn account	30.5%	27.1 – 34.1
Balance above 2,000	57.8%	55.8 – 59.8

- Proportions use Wilson intervals. The gap between no housing loan and housing loan is **20.4 points (95% CI 18.6 – 22.2)**.
- Mean call duration 370.5 s (6.2 minutes), CI 364 – 377; subscribers 534 s (523 – 544) vs non-subscribers 223 s (218 – 229), non-overlapping.
- Median balance 551 (bootstrap CI 542 – 558); the mean (1,520, CI 1,460 – 1,579) is inflated by a few large accounts.
- Proxy cost per subscriber (assuming 0.50 per call-minute): 6.52 (6.40 – 6.65). This reflects sampling noise only, not whether 0.50 is the right price.
- Channel ROI (bootstrap, campaigns resampled): Paid Search 228% (154 – 323), Referral 295% (176 – 473), Email 313% (142 – 619), PPC 353% (215 – 564), Social Media 379% (221 – 604). The intervals overlap heavily; Social Media beat Paid Search in 94% of resamples, but that pair was picked after seeing the data, so it is suggestive only.

Interpretation: the headline gaps (previous success, unknown contact, housing loan, many calls) are far larger than their sampling error. Small segments carry wider intervals and should be quoted with them.

5. Feature engineering and evaluation

Eleven features were engineered, each with a stated logic and business purpose (see the Feature Documentation): age_group, balance_tier, balance_log, debt_status, contact_frequency_band, previously_contacted, previous_success, contact_channel_known, quarter, day_period and call_duration_band (post-call only).

Before vs after

- **Signal visibility:** age alone has |r| = 0.04 with the outcome but age_group reaches V = 0.20; balance_tier V = 0.145 vs |r| = 0.08 for raw balance; contact_frequency_band V = 0.14 vs |r| = 0.13.
- **Ranking quality** (pre-call information only, 10 random 70/30 splits): AUC 0.745 with the 9 original categorical columns vs about 0.751 after adding the pre-call engineered features, a gain of roughly 0.006 (split-to-split spread about 0.008). Adding the post-call duration band lifts AUC to 0.84, which shows how much is decided during the call. The top 30% of ranked customers capture 48.6% of subscribers before vs 49.0% after.
- **Segmentation** (built on training data, judged on unseen customers): the 10 best age × debt status × previously-contacted segments cover 24.8% of customers, respond at 74.5% and capture 39.1% of subscribers; the best jobs cover 28.8%, respond at 57.0% and capture 34.7%.

Verdict: feature engineering gave a modest ranking gain but clearly better, easier-to-explain customer segments.

Feature	Decision
age_group, balance_tier, contact_frequency_band	Keep (each adds signal)
previous_success, previously_contacted	Keep for segments (poutcome already holds the information)
contact_channel_known	Keep as a data-quality flag
debt_status	Keep for segments; redundant with housing + loan inside a model
balance_log	Keep for correlation or regression use (skew 8.28 → −1.81)

Feature	Decision
quarter, day_period	Drop (no added value; weaker than month)
call_duration_band	Post-call diagnostic only

6. Limitations

- Observational data: associations do not prove that changing a factor changes the response.
- Exploration and testing used the same customers; the hold-out evaluation is the safeguard.
- Month, contact method and campaign intensity move together and cannot be separated here.
- The ranking model in section 5 is a simple comparison tool, not a production scoring model.
- ROI inputs (0.50 per minute, 200 per deposit) are assumptions; the campaign table is small (88 rows) and unlinked to customers.